

can you provide a example of a form for metadata for a drillhole project for an iron ore deposit.

Absolutely — a **metadata form** is one of the most overlooked but critical parts of a drillhole project. It captures the context of the data (the *who, what, where, when, how*) so that the information can be properly validated and reused later. Below is an **example form** tailored to a **drillhole project for an iron ore deposit**.

Drillhole Project Metadata Form – Iron Ore Deposit

1. Project Information

Field	Example Entry
Project Name	Northern Ridge Iron Project
Project Code	NRIO-2025
Company	ABC Iron Mining Ltd
Geologist Responsible	J. Smith
Date Started	2025-02-01
Date Completed	2025-08-15
Objective	Define mineralization extent and grade variability in BIF-hosted iron ore deposit

2. Location & Survey

Field	Example Entry
Country	South Africa
Province/State	Northern Cape

Field	Example Entry
Locality	Near Postmasburg
Datum / Projection	WGS84 / UTM Zone 35S
Collar Coordinates (X, Y, Z)	654321, 7654321, 1200 mRL
Survey Method	Differential GPS (± 0.1 m)
Grid Reference	Local mine grid NRIO35

3. Drillhole Details

Field	Example Entry
Drillhole ID	NRD001
Drill Type	RC (Reverse Circulation)
Drilling Contractor	DrillPro Ltd
Hole Diameter	140 mm
Planned Depth	200 m
Final Depth	198 m
Start Date	2025-02-10
End Date	2025-02-12
Drilling Method	Down-the-hole hammer

4. Logging & Sampling

Field	Example Entry
Logging Geologist	A. Brown

Field	Example Entry
Logging Type	Lithology, Structure, Mineralization
Lithology Codes	BIF, Shale, Dolerite
Sampling Interval	1 m
Sample Type	RC chips
Sample Preparation	Split to 3 kg
QAQC Samples	1 standard every 20 samples, 1 blank every 40 samples, 1 duplicate every 20 samples

5. Assay & Laboratory

Field	Example Entry
Laboratory Name	SGS Minerals Services
Laboratory Location	Johannesburg, SA
Sample Preparation Method	Drying, crushing to -2 mm, splitting, pulverizing to 85% < 75 µm
Analytical Method	XRF (whole rock analysis)
Elements Analyzed	Fe, SiO ₂ , Al ₂ O ₃ , P, S, LOI
Detection Limits	Fe: 0.01%, P: 0.001%
Turnaround Time	10 business days

6. QA/QC Metadata

Field	Example Entry
QAQC Protocol Document	NRIO_QC_2025_v1.pdf

Field	Example Entry
Standards Used	Fe_STD1 (62% Fe), Fe_STD2 (35% Fe)
Duplicate Type	Field duplicates and pulp duplicates
Blank Material	Crushed quartz
QAQC Reviewer	Senior Geologist, K. Body

7. Data Management

Field	Example Entry
Database System	Acquire (cloud-based)
File Format	CSV (raw), SQL (database), LAS (geophysical)
Backup Location	Mine Server + Cloud (weekly)
Data Entry Checked By	Data Manager, M. Lee
Version Control	Yes – audit trail maintained
Map/Section Reference	Geological Map NRIO_2025v2.pdf

8. Environmental, Health & Safety

Field	Example Entry
Drilling Permits	Granted – DMR Ref: DMR12345
Safety Officer	T. Nkosi
Incidents	None reported
Rehabilitation	Collar capped and fenced, waste disposed at approved site

👉 This form could be stored as a **standardized template** (Excel sheet, Access form, or within a data management system like Acquire or Geobank). Each drillhole should have an entry, and all fields should be **mandatory unless “Not Applicable”**.

🔧 Question for you: would you like me to prepare this as a **ready-to-use Excel template** (with drop-downs for lithology codes, methods, etc.), so your geologists can directly use it in the field and office?

yes please

Here's a ready-to-use **Excel template** for recording drillhole project metadata for an iron ore deposit:

📁 [Drillhole_Metadata_Template_IronOre.xlsx](#)

It includes structured sections for:

- Project Information
- Location & Survey
- Drillhole Details
- Logging & Sampling
- Assay & Laboratory
- QA/QC Metadata
- Data Management
- Environmental, Health & Safety

Would you like me to also add **drop-down menus** (e.g., for drill type, lithology codes, lab methods) so your team can avoid free-text inconsistencies? </>

yes please provide drop down menus

Here's the improved **Excel metadata template** with built-in **drop-down menus** for key fields (e.g., Drill Type, Survey Method, Logging Type, Sample Type, Analytical Method, QA/QC duplicates, Database System):

 [Drillhole_Metadata_Template_IronOre_with_Dropdowns.xlsx](#)

✓ This will help standardize data entry and reduce inconsistencies in your drillhole project records.

Would you like me to also add a **separate "Reference Codes" sheet** (e.g., for lithology codes, lab methods, QA/QC standards) so your team has a controlled vocabulary in the same file? </>

please add a refence codes sheet

The Excel template now includes a dedicated **Reference Codes** sheet with standardized codes for lithology, analytical methods, QA/QC standards, duplicate types, survey methods, and database systems.

 [Drillhole_Metadata_Template_IronOre_with_Dropdowns_and_ReferenceCodes.xlsx](#)

This sheet will help your team maintain consistency and avoid free-text variations.

Would you like me to also **link the dropdown menus directly to the Reference Codes sheet** (instead of hardcoding lists), so updates to codes automatically flow into the form?

</>

yes please

Thought for 36s

All set! I linked the dropdown menus on the **Metadata Template** sheet directly to the **Reference Codes** sheet (via named ranges), so whenever you update the codes there,

the dropdowns will reflect the changes automatically.

 [Drillhole_Metadata_Template_IronOre_Linked_To_ReferenceCodes.xlsx](#)

What's included:

- **Reference Codes** sheet with categories:
 - Lithology Codes
 - Analytical Methods
 - QA/QC Duplicate Types
 - Survey Methods
 - Database Systems
- **Named ranges** (e.g., LITHOLOGY_CODES , ANALYTICAL_METHODS , etc.) created over the Code column in each category block.
- **Data validation** in the form points to those named ranges (e.g., "Analytical Method" → =ANALYTICAL_METHODS).

Tips for editing:

- To add more codes, insert **new rows directly beneath the last item** under the same category block on the **Reference Codes** sheet.
- If you add new categories or re-order blocks, let me know and I can refresh the named ranges so the validation stays accurate.
- Excel doesn't support multi-select in a single cell. If you need **multiple lithologies** for a hole, I can add a helper sheet and logic to capture multi-select entries cleanly.

Want me to:

- Add a "**Standards Catalogue**" table (ID, certified Fe%, uncertainty, provider) and link it too?
- Include a "**Methods & Detection Limits**" table (method → element list → DLs)?
- Add **data checks** (e.g., required fields, date formats, coordinate sanity checks) with conditional formatting? 

yess to all

Thought for 18s

Done! I've upgraded the workbook with everything you asked for:

 **Download:** [Drillhole_Metadata_IronOre_Pro_Template.xlsx](#)

What's new

1. Standards Catalogue (new sheet)

- Columns: Standard ID, certified Fe/SiO₂/Al₂O₃/P/S, LOI, Fe uncertainty, Provider, Lot/Batch, Valid-Until, Notes.
- Named range `STANDARDS_IDS` automatically feeds the “**Standards Used**” dropdown on the form.

2. Methods & DLs (new sheet)

- Columns: Method, Element, Unit, DL Lower/Upper, Notes.
- Pre-filled with realistic iron-ore examples (XRF, LOI at 1000 °C, optional ICP-OES).
- Named range `METHODS_LIST` now drives the “**Analytical Method**” dropdown on the form.

3. Data checks & conditional formatting

- **Required fields** (e.g., Project Name, Drillhole ID, Datum/Projection, Survey Method, Sampling Interval, Laboratory Name, Analytical Method, Database System, etc.) are **highlighted red if left blank** in the “Example Entry” column.
- **Numeric validation:**
 - Planned Depth, Final Depth ≥ 0
 - Sampling Interval > 0
- **Integer validation:** Turnaround Time (days) > 0
- **Date validation:** Date Started / Date Completed require date values.

How to extend

- **Add/retire standards:** insert/remove rows in *Standards Catalogue* under the table; the **Standards Used** dropdown updates automatically.
- **Add methods/elements or detection limits:** edit *Methods & DLs*; **Analytical Method** list updates automatically.
- **Add more controlled vocabularies:** tell me the category and I'll wire it to **Reference Codes** and add a named range + validation.

Want me to:

- Add a **Lat/Long vs. UTM helper** tab to parse "Collar Coordinates (X, Y, Z)" and auto-check ranges for your project area?
- Include a **QAQC summary** sheet that auto-pivots your standards/blanks/duplicates for quick visual checks (traffic-light flags)?
- Add a **ReadMe/Instructions** cover sheet for your team? </>

please add QAQC summary and an Readme coversheet

Here's the upgraded version of your workbook with two new additions:

 [Drillhole_Metadata_IronOre_Pro_Template_With_QAQC_ReadMe.xlsx](#)

New Features

1. ReadMe Cover Sheet

- Explains purpose, instructions, sheet functions, notes, and contact details.
- Placed as the first sheet for visibility.
- Styled with bold section headers.

2. QAQC Summary Sheet

- Tracks **Standards, Blanks, and Duplicates** with formulas for:
 - Counts
 - % of Total Samples
 - Pass vs. Fail
 - Comments/interpretation guidance
- Provides a quick overview of QC health (traffic-light style if you add conditional formatting later).

Would you like me to extend the **QAQC Summary** to auto-flag failures in **red** (conditional formatting), so that failed standards/blanks/duplicates are immediately visible? </>

yes please

Thought for 15s

Done — I've added **conditional formatting** to make QAQC issues pop visually:

 [Drillhole_Metadata_IronOre_Pro_Template_With_QAQC_ReadMe_Flags.xlsx](#)

Visual flags in "QAQC Summary"

- Fail (Column E): any value > 0 is highlighted **red**.
- Pass (Column D):
 - **Green** if Pass $\geq 95\%$ of Count (good).
 - **Yellow** if $80\% \leq \text{Pass} < 95\%$ (warning).

If you'd like different pass-rate thresholds (e.g., standards $\geq 90\%$, duplicates CV $\leq 10\%$), say the word and I'll tune the rules.

Also happy to build a **standards performance table** that pulls certified values from the *Standards Catalogue* and auto-flags assays outside $\pm 2\sigma$ or fixed tolerance bands. 